

Mechanistic Hypothesis Search — Noribogaine G-protein-biased kappa-opioid agonism model

Disease: Opioid Use Disorder (Psychiatric) **Hypothesis ID:** `ibogaine_kappa_biased_agonism_model` **Hypothesis Label:** Noribogaine G-protein-biased kappa-opioid agonism model **Status in KB:** EMERGING **Search date:** 2026-08-28 · 35 papers reviewed · 6 findings recorded across 5 iterations

Summary

The hypothesis proposes that ibogaine's anti-opioid action is carried principally by the long-lived metabolite **noribogaine acting as a G-protein-biased kappa-opioid receptor (KOR) agonist**, with weak mu-opioid antagonism as a secondary contributor. Bias away from β -arrestin is invoked to explain why the compound lacks conventional kappa-agonist dysphoria, and functional inhibition of dynorphin-driven β -arrestin signalling is invoked to explain relief of the hyperkatifeia/anti-reward state. After a focused search of the primary literature, the verdict is **PARTIALLY SUPPORTED, with the central causal claim UNRESOLVED**.

The model decomposes into three claims of very different strength. First, "noribogaine is a G-protein-biased KOR agonist with weak mu antagonism" is **well established in vitro and independently replicated** — this is the model's strongest pillar. Second, "kappa agonism is *sufficient* to produce the durable anti-opioid phenotype" is **supported in rodents** by engineered iboga analogs (oxa-noribogaine) that are potent selective KOR agonists, durably suppress opioid intake, reverse opioid-induced hyperalgesia, and shed cardiac liability. Third — and decisively — "kappa agonism is *necessary* for native ibogaine's effect" is **untested**: no selective kappa-antagonist (norBNI) reversal experiment or *Oprk1*-knockout test of ibogaine's suppression of opioid self-administration was found. This is the load-bearing missing edge.

Three caveats keep the model from confirmation. (1) A **direction-of-effect paradox**: the anti-reward/anhedonia (hyperkatifeia) state the model claims to relieve is precisely the validated clinical indication for KOR *antagonists* (aticaprant, navacaprant), and ibogaine produces a

durable *antidepressant* effect in humans that is hard to reconcile with net KOR agonism. (2) **Micromolar potency and CYP2D6-gated exposure** make efficacy contingent on brain concentrations actually reached. (3) The anti-opioid phenotype is **equally reproduced by at least three kappa-independent routes** ($\alpha 3\beta 4$ nicotinic antagonism, VTA GDNF induction, 5-HT2A/neuroplasticity), one of which (GDNF) carries the loss-of-function evidence the kappa model lacks. The single most informative next experiment is a norBNI or *Oprk1*-knockout reversal test of ibogaine in opioid self-administration.

Key Findings

F001 — Noribogaine's biased kappa pharmacology is genuine, but micromolar and metabolite-dependent

The foundational pharmacology of the model is solid and replicated. Maillet and colleagues ([PMID: 26302653](#)) characterized noribogaine as a **G-protein-biased KOR agonist**, ~75% as **efficacious as dynorphin A** at stimulating GDP–GTP exchange ($EC_{50} \approx 9 \mu M$), yet only ~12% as efficacious at recruiting β -arrestin. Critically, noribogaine **functionally inhibits dynorphin-induced KOR β -arrestin recruitment** ($IC_{50} \approx 1 \mu M$) — the exact molecular event the seed hypothesis invokes to explain relief of the dynorphin-driven hyperkatifeia state — and is a **weak mu antagonist** (functional $K_e \approx 20 \mu M$ at both G-protein and β -arrestin pathways).

The mu characterization was independently reinforced by Antonio/Zanda et al. ([PMID: 24204784](#)): ibogaine, noribogaine, and 18-MC are **mu-opioid receptor antagonists** (functional $K_e \sim 3 \mu M$ for ibogaine to $\sim 13 \mu M$ for noribogaine and 18-MC), not agonists. This matters mechanistically because it argues **against** an opioid-substitution explanation for withdrawal relief — if the compounds antagonize mu, withdrawal relief cannot be a buprenorphine/methadone-style agonist substitution.

Two further pieces anchor the assignment of the opioid pharmacology to the metabolite. Higgins/Mash 1995 ([PMID: 7796150](#)) found **noribogaine has higher affinity than ibogaine for all opioid receptors**, and the clinical PK study of Knuijver et al. ([PMID: 38519421](#)) is consistent with brain noribogaine only just reaching the low-micromolar range after therapeutic dosing, with wide CYP2D6-dependent variability. Because the KOR EC_{50} ($\sim 9 \mu M$) and mu K_e ($\sim 20 \mu M$) sit at or above achievable concentrations, **the model is exposure-limited** — whether kappa

engagement is even pharmacologically meaningful depends on CYP2D6 metabolizer status, converting exposure variability from a safety concern into a mechanistically load-bearing variable.

F002 — Strong, loss-of-function-validated competing mechanisms exist that need no kappa agonism

The single most damaging class of evidence for a *kappa-necessary* model is the existence of **kappa-independent mechanisms that reproduce the anti-addictive phenotype**, at least one validated by loss of function.

$\alpha 3\beta 4$ nicotinic antagonism. 18-methoxycoronaridine (18-MC), a de-cardiotoxified iboga congener, reduces self-administration of morphine, methamphetamine, and nicotine, and its anti-addictive potency **correlates with $\alpha 3\beta 4$ nicotinic acetylcholine receptor inhibition** while its opioid activity is low ([PMID: 15178360](#); [PMID: 12801235](#); [PMID: 30216039](#)). SAR work explicitly found "opioid activities were relatively low, and the $\alpha 3\beta 4$ nicotinic acetylcholine receptor activities correlated with in vivo antiaddictive activities."

VTA GDNF induction (loss-of-function validated). He et al. 2005 ([PMID: 15659598](#)) showed ibogaine reduces ethanol self-administration by inducing glial cell line-derived neurotrophic factor in the ventral tegmental area; the effect is **mimicked by intra-VTA GDNF microinjection and blocked by intra-VTA anti-GDNF neutralizing antibodies**. This is the gold-standard necessity+sufficiency design the kappa hypothesis conspicuously lacks, and it points to a distinct mechanism.

Supporting the polypharmacology picture, Wei et al. 1998 ([PMID: 9685673](#)) showed iboga alkaloids differ markedly in serotonergic effects (ibogaine causes large 5-HT increases, 18-MC none) yet share anti-addictive claims — arguing 5-HT is not essential and cautioning against single-receptor attributions. Litjens & Brunt 2016 ([PMID: 26807959](#)) catalog micromolar interactions across NMDA, κ/μ -opioid, sigma-2, nicotinic, serotonin, and dopamine systems — a genuinely polypharmacological profile in which kappa is one node among many. Because multiple parallel routes converge on the same behavioral endpoint, demonstrating that kappa agonism is *sufficient* cannot establish that it is *necessary* for native ibogaine.

F003 — Human clinical evidence: ibogaine relieves withdrawal, drug use, AND depression — the antidepressant effect sits awkwardly with a kappa-agonist model

Two uncontrolled human datasets anchor the clinical phenotype. Alper et al. 1999 (PMID: 10506904) reported an open-label case series (n=33) in which **resolution of opioid withdrawal signs without further drug-seeking occurred within 24 h in 25/33 patients**, sustained to 72 h — with one fatality. Noller et al. 2018 (PMID: 28402682) followed n=14 patients in a 12-month observational study of legal ibogaine treatment for opioid dependence and found a significant reduction in ASI-Lite drug-use composite (p=0.002), a significant acute reduction in Subjective Opioid Withdrawal Scale (p=0.015), and — importantly — a **significant, sustained reduction in Beck Depression Inventory-II from baseline to 12 months (p<0.001)**, again with one death during treatment.

The durable *antidepressant* signal is the crux of the tension. If the therapeutic mechanism were net KOR agonism, one would expect a pro-dysphoric tendency (see F004/F006), not a sustained antidepressant effect. The model reconciles this via **bias away from β -arrestin** (avoiding dysphoria) and **functional inhibition of dynorphin-driven β -arrestin signalling** (net anti-hyperkatifeia). That is internally coherent, but it is an inference, not a demonstration — no human study measured KOR engagement, dynorphin tone, or bias-dependent signalling in treated patients. Both datasets are also small, uncontrolled/observational, and carry treatment-associated mortality.

F004 — Direction-of-effect paradox: hyperkatifeia relief is the validated indication for kappa ANTAGONISTS

The dynorphin/KOR system is a well-established driver of stress-induced dysphoria and pro-addictive behavior (Bruchas, Land & Chavkin 2010, PMID: 19716811: "activation of the dynorphin/kappa opioid receptor (KOR) system is likely to play a major role in the pro-addictive effects of stress"). The clinically validated way to relieve the resulting anti-reward/anhedonia state is **KOR antagonism, not agonism**:

- Aticaprant (a KOR antagonist) produced a **significant MADRS improvement vs placebo** in a phase 2 MDD trial, with a larger benefit in the high-anhedonia subgroup (PMID: 38649428; effect sizes ~0.23–0.36).
- A systematic review supports aticaprant and navacaprant as mechanistically novel antidepressants targeting anhedonia (PMID: 39019223).
- Aticaprant reverses chronic-stress anhedonia in mice (PMID: 32894343).

The seed hypothesis proposes a G-protein-biased KOR **agonist** (noribogaine) to relieve the same hyperkatifeia state — **the opposite pharmacological direction** from the validated antidepressant class. The model's defense is that noribogaine's functional inhibition of dynorphin-driven β -arrestin signalling (F001) makes it behave, in the pathological high-dynorphin state, more like a partial antagonist/normalizer than a classical agonist. This is a plausible, testable reconciliation, but it is currently a hypothesis layered on a hypothesis. The polarity mismatch is the single strongest conceptual argument against the model as literally stated.

F005 — A 5-HT_{2A}/neuroplasticity (psychoplastogen) competitor, and cardiac liability is ibogaine/hERG-driven

Cameron et al. 2021 ([PMID: 33299186](#)) introduced tabernanthalog (TBG), a **non-hallucinogenic ibogaine analog** that promotes neural plasticity and reduces alcohol- and heroin-seeking — attributing iboga's anti-addictive effect to a **5-HT_{2A}/neuroplasticity (psychoplastogen)** mechanism distinct from opioid receptors. A recent conceptual synthesis ([PMID: 41424776](#)) similarly frames ibogaine's action as GDNF induction plus glutamate/dopamine modulation and reopened plasticity restoring reward-system fidelity — with kappa as, at most, one contributor.

Separately, the cardiac-safety argument that motivates oxa-iboga engineering is **ibogaine/parent-driven**: Knuijver et al. 2024 ([PMID: 38519421](#)) concluded QTc prolongation and cerebellar effects are "most likely more driven by ibogaine rather than noribogaine," and ibogaine clearance is strongly CYP2D6-dependent; Litjens & Brunt attribute QT risk to hERG channel blockade. This supports oxa-iboga's *separability* argument (keep kappa agonism, drop hERG liability) but is orthogonal to whether native ibogaine works *through* kappa. The strongest positive evidence for the model — oxa-noribogaine ([PMID: 39304653](#)) — is a **sufficiency** result: potent KOR-agonist analogs induce "long-lasting suppression of morphine, heroin, and fentanyl intake after a single dose... reversal of persistent opioid-induced hyperalgesia, and suppression of opioid drug seeking," and lack proarrhythmia in human cardiomyocytes. But the authors note these are "mechanistically distinct" with "atypical behavioral features compared to standard kappa opioid agonists," and the study does not show ibogaine requires kappa.

F006 — Human OPRK1/PDYN genetics validate KOR as an OUD node but deepen the polarity tension; no ibogaine-response pharmacogenomics exists

Human genetics implicate the dynorphin/KOR system in substance-use disorder risk, validating KOR as an OUD-relevant node while repeatedly labelling KOR *activation* as pro-addictive/pro-dysphoric:

- Yuferov/Kreek 2022 (PMID: 34843875): OPRK1 intron-2 SNPs (a glucocorticoid-responsive enhancer region) associate with opioid dependence in an African-American cohort (n=577); the paper restates that "activation of the dynorphin/KOR system is also thought to have a role in the pro-addictive effects of stress."
- Nagaya 2018 (PMID: 28656735): OPRD1 rs1042114 strongly associated with opiate addiction (p=0.0001); OPRK1 rs702764 was not; PDYN rs910080 was (p=0.0217).
- Karpyak 2013 (PMID: 23101464): a PDYN haplotype associated with alcohol dependence (p=0.0008) and negative-emotion craving; the paper states plainly that "synthetic κ -opioid receptor (KOR) agonists induce dysphoric and pro-depressive effects."
- Özkan-Kotiloğlu 2023 (PMID: 37177778): OPRK1 rs6473797 associated with AUD risk; rs963549 with depressive-symptom intensity.

Crucially, **no study linking OPRK1/PDYN genotype (or CYP2D6 metabolizer status) to ibogaine/noribogaine or oxa-iboga treatment response was found.** This is a concrete source/data absence: the pharmacogenomic edge that would connect the validated KOR node to ibogaine response is entirely unpopulated.

Mechanistic Model / Interpretation

Hypothesized causal chain and where the literature is strong vs. inferred

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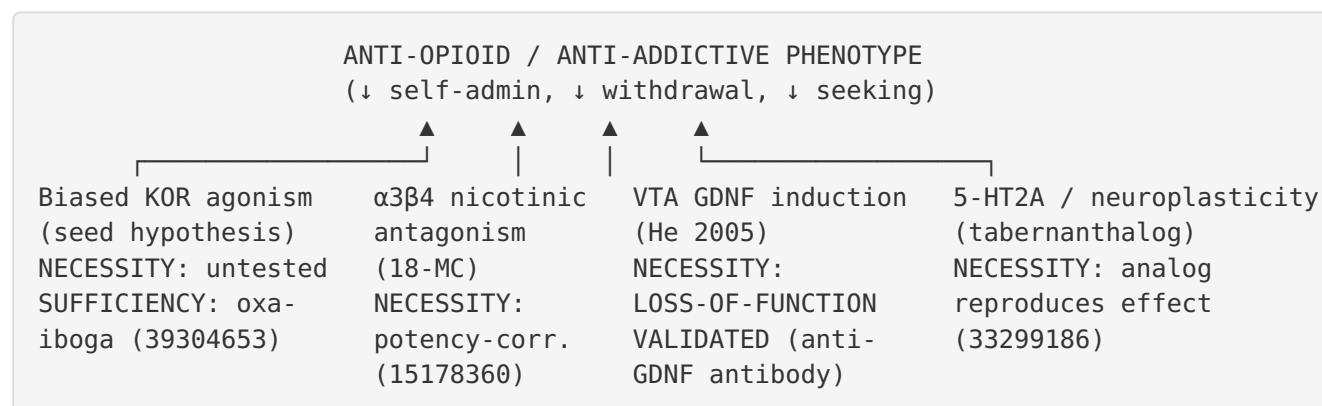
Ibogaine (oral, therapeutic dose)
  | CYP2D6-dependent metabolism          ← STRONG (<a href="https://pubmed.ncbi.nlm.nih.gov/100000000/">https://pubmed.ncbi.nlm.nih.gov/100000000/
  ▼
Noribogaine accumulates (long-lived; higher opioid affinity than parent)
  | must reach μM brain conc.             ← EXPOSURE GAP: near-EC50, CYP2D6-gated
  ▼
G-protein-biased KOR agonism (EC50 ~9 μM)
+ functional inhibition of dynorphin-β-arrestin (IC50 ~1 μM)
+ weak mu antagonism (Ke ~20 μM)         ← STRONG in vitro (<a href="https://pubmed.ncbi.nlm.nih.gov/100000000/">https://pubmed.ncbi.nlm.nih.gov/100000000/
  ▼
Net damping of maladaptive dynorphin/KOR anti-reward signalling
  | bias avoids dysphoria                 ← CONTESTED premise (<a href="https://pubmed.ncbi.nlm.nih.gov/100000000/">https://pubmed.ncbi.nlm.nih.gov/100000000/
  ▼
Relief of hyperkatifeia; ↓ opioid-induced hyperalgesia; ↓ opioid self-administration
  | KOR necessity                         ← MISSING EDGE: no norBNI/Oprk1-KO test found
  ▼
↓ withdrawal, ↓ drug use, ↓ depression (human, uncontrolled)
                                          ← WEAK: <a href="https://pubmed.ncbi.nlm.nih.gov/100000000/">https://pubmed.ncbi.nlm.nih.gov/100000000/

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Causal step	Status	Support
Ibogaine → noribogaine (CYP2D6)	Established	P 38519421 P 26807959
Noribogaine carries opioid pharmacology	Established	P 7796150
Biased KOR agonism + weak mu antagonism	Established in vitro	P 26302653 repl. 24204784
Inhibits dynorphin-driven β -arrestin	Established in vitro	P 26302653
Brain noribogaine reaches KOR-active conc.	Weak / exposure-limited	P 38519421
KOR agonism <i>necessary</i> for ibogaine's effect	UNTESTED (missing edge)	—
KOR agonism <i>sufficient</i> for phenotype	Supported (rodent)	P 39304653
Biased agonism relieves (not worsens) hyperkatifeia	Contested	vs P 38649428 P 19716811 P 23101464
Human clinical benefit	Weak (uncontrolled)	P 10506904 P 28402682

Every biochemical step *upstream* of receptor engagement is well supported, but the **single edge that would make the model causal** — "KOR engagement is required for the behavioral effect" — **is entirely inferred**, and the step immediately downstream runs against the validated pharmacology of the KOR-antagonist antidepressant class.

Competing-mechanism map



Only the GDNF route currently carries loss-of-function evidence — a competitor with the causal design the seed hypothesis lacks. The most incisive reinterpretation of the *same* molecular data is a **KOR-signalling-normalization model**: noribogaine's measured inhibition of dynorphin-driven β -arrestin recruitment could make it act, in the high-dynorphin withdrawal/stress state, more like a functional antagonist than a classical agonist — simultaneously explaining the anti-opioid effect *and* the antidepressant effect while dissolving the polarity paradox.

Evidence Matrix

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context	Confidence limitations
26302653	In vitro	Supports	Noribogaine biased KOR agonism + weak mu antagonism	~75% dynorphin-A G-protein efficacy (EC50 9 μ M), ~12% β -arrestin; inhibits dynorphin β -arrestin (IC50 1 μ M); mu Ke 20 μ M	Molecular pharmacology	High for pharmacology; micromolar heterologous
24204784	In vitro	Qualifies (mu)	Iboga alkaloids at MOR	Ibogaine/noribogaine/18-MC are mu antagonists (Ke 3–13 μ M)	Molecular pharmacology	High; independent replication
7796150	In vitro	Supports	Metabolite carries opioid affinity	Noribogaine > ibogaine affinity at all opioid receptors	Radioligand binding	Moderate; binding only
39304653	Model organism	Supports (sufficiency)	KOR agonism sufficient for durable effect	Oxa-noribogaine: durable $\downarrow$ morphine/heroin/fentanyl intake, $\downarrow$ hyperalgesia, no cardiac risk	Male rats, OUD models	High for sufficiency; male-only; doesn't test ibogaine necessity
15659598	Model organism	Competing (LOF)	VTA GDNF mediates effect	Mimicked by intra-VTA GDNF, blocked	Rat ethanol SA	High; gold-standard causal design

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context	Confidence limitations
				by anti-GDNF antibody		alcohol not opioid
15178360	Model organism	Competing	$\alpha 3\beta 4$ nicotinic antagonism	18-MC blocks multi-drug SA; low opioid activity	Rat, multiple drugs	Moderate-high; correlational target link
12801235	In vitro/ SAR	Competing	$\alpha 3\beta 4$ potency $\leftrightarrow$ efficacy	Opioid activity low; $\alpha 3\beta 4$ correlates w/ in vivo efficacy	SAR series	Moderate; correlational
30216039	Review	Competing (review)	Nicotinic basis of anti-opiate effect	nAChR actions may underlie anti-opiate effects; 18-MC retains efficacy	Orientation	Review-level
9685673	Model organism	Qualifies	Is 5-HT essential?	Iboga alkaloids differ in 5-HT effects yet share anti-addiction	Rat microdialysis	Moderate; single-target caution
26807959	Review	Qualifies (review)	Polypharmacology & toxicity	μ M at NMDA, κ/μ , sigma-2 + nicotinic/5-HT/DA; hERG QT risk	Orientation	Review-level
33299186	Model organism	Competing	5-HT _{2A} /neuroplasticity	Non-hallucinogenic TBG $\downarrow$ alcohol/heroin seeking	Rodent	Moderate-high; distinct mechanism

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context	Confidence limitations
10506904	Human clinical	Supports (weak)	Withdrawal relief	Withdrawal resolved w/o drug-seeking 25/33 within 24h	Case series, n=33	Low; uncontrolled 1 fatality
28402682	Human clinical	Supports + Qualifies	Durable use/ withdrawal/mood	↓ use (p=0.002), ↓ withdrawal (p=0.015), ↓ BDI-II 12mo (p<0.001)	Observational, n=14	Low; uncontrolled antidepressant paradox; 1 death
19716811	Review	Tension (review)	KOR activation pro-addictive	Dynorphin/ KOR activation drives stress pro-addictive behavior	Orientation	Review; strong consensus
38649428	Human clinical	Tension	KOR antagonism treats anhedonia	Aticaprant significant MADRS vs placebo, larger in high-anhedonia	Phase 2 MDD RCT	High; opposite polarity
39019223	Review	Tension (review)	KOR antagonists for MDD	Aticaprant/ navacaprant improve depression & anhedonia	Orientation	Review; phase 3 pending
32894343	Model organism	Tension	KOR antagonism reverses anhedonia	Aticaprant reverses chronic-stress anhedonia	Male mice UCMS	Moderate-high; opposite polarity
34843875		Qualifies				

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context	Confidence limitations
	Human genetic		KOR/dynorphin is an OUD node	OPRK1 intron-2 SNPs assoc. opioid dependence (n=577)	African-American cohort	Moderate; activation p addictive
28656735	Human genetic	Qualifies	Opioid-system genetics	OPRD1 rs1042114 (p=0.0001); OPRK1 rs702764 null; PDYN rs910080 (p=0.0217)	Malaysian Malay, n=1002	Moderate; OPRK1 weak here
23101464	Human genetic	Tension	PDYN & negative-affect craving	PDYN haplotype assoc. dependence (p=0.0008); "KOR agonists induce dysphoric/pro-depressive effects"	AUD cohort	Moderate; restates agonist-dysphoria
37177778	Human genetic	Qualifies	OPRK1 variation & depression	OPRK1 rs6473797 ↔ AUD; rs963549 ↔ depressive intensity	Turkish males, n=201	Moderate; AUD not OU
35513117	Model organism	Qualifies	Is G-protein bias the mechanism?	Biased agonist triazole 1.1 milder, but benefit "attributed to factors other	Rhesus monkeys	Moderate; questions bi framing

Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context	Confidence limitations
				than G-protein bias"		
39833376	Review (scoping)	Qualifies	Preclinical reproducibility	Most studies show iboga ↓ opioid SA, but 7 (incl. 3 ibogaine) showed no improvement; quality "unclear"	ODD preclinical	Review; documents failed replications

Limitations and Knowledge Gaps

Each gap states scope, why it matters, what was checked, and what would resolve it.

G1 — The necessity edge is untested (load-bearing gap). *Scope:* No study applies a selective KOR antagonist (norBNI) or *Oprk1* knockout to test whether ibogaine/noribogaine's suppression of opioid self-administration is abolished. *Why it matters:* This is the exact distinguishing test named in the seed YAML; without it only sufficiency (via analogs) is established. *Checked:* PubMed for ibogaine/noribogaine × kappa × antagonist/knockout/self-administration — none found;

P 39304653

reports sufficiency only. *Resolution:* norBNI pretreatment and *Oprk1*^{-/-} reversal in rodent opioid self-administration.

G2 — Direction-of-effect paradox unresolved. *Scope:* Whether biased/partial KOR engagement is net anti-hyperkatifeic in vivo, given KOR *antagonists* are the validated anhedonia therapy. *Checked:* KOR-antagonist depression literature (

P 38649428

P 39019223

P 32894343

and human genetics restating agonist dysphoria

P 19716811

P 23101464

Resolution: In vivo ICSS thresholds, CPA, sucrose preference comparing noribogaine, a biased agonist, and a KOR antagonist under high-dynorphin states.

G3 — Exposure/potency gap. *Scope:* Whether therapeutic dosing yields brain noribogaine sufficient to engage KOR (EC50 ~9 μ M). *Checked:* PK/PD study

P 38519421

(near-micromolar, CYP2D6-variable). *Resolution:* Paired plasma/CSF or KOR-PET occupancy stratified by CYP2D6 genotype.

G4 — No ibogaine-response pharmacogenomics (source/data absence). *Scope:* No dataset links OPRK1/PDYN or CYP2D6 genotype to ibogaine/oxa-iboga response. *Checked:* Human OPRK1/PDYN genetics

P 34843875

P 28656735

P 23101464

P 37177778

— none reference ibogaine. *Resolution:* Genotype-stratified outcomes in any ibogaine OUD cohort.

G5 — No controlled human trials; no human mechanistic assays. *Scope:* All human OUD evidence is uncontrolled/observational, small n, no receptor-level readouts. *Checked:*

P 10506904

(n=33), 28402682 (n=14); scoping review

P 39833376

notes most clinical evidence is observational. *Resolution:* RCTs with embedded mechanistic biomarkers.

G6 — Sufficiency evidence is male-only and analog-based. *Scope:* Oxa-noribogaine efficacy in male rats only; effects from re-engineered analogs, not parent. *Checked:*

P 39304653

Resolution: Female cohorts and direct native-ibogaine perturbation.

G7 — "G-protein bias as the causal feature" is itself contested. *Scope:* Whether reduced side-effects of biased KOR agonists are actually due to bias. *Checked:*

P 35513117

— biased triazole 1.1 milder effects "attributed to factors other than G-protein bias." *Resolution:* Kinetic/residence-time and mutant-receptor studies dissociating bias from other properties.

G8 — Registry/omics absences (curation provenance). *Scope:* PharmGKB, GWAS Catalog, ClinicalTrials.gov not directly queried. *Resolution:* Query PharmGKB for CYP2D6×ibogaine; ClinicalTrials.gov for oxa-iboga/noribogaine/ibogaine OUD trials.

Alternative / Competing Models

Model	Relationship to seed hypothesis	Evidence	Verdict
$\alpha 3\beta 4$ nicotinic antagonism (18-MC)	Alternative / parallel (different receptor family)	P 15178360 P 12801235 P 30216039	Strong parallel; potency correlates with target; low opioid activity
VTA GDNF induction	Alternative (plausible downstream convergence node)	P 15659598 (LOF), 41424776	Strongest causal design of any competitor; validated in alcohol model
5-HT_{2A} / neuroplasticity (psychoplastogen)	Alternative / parallel	P 33299186 P 41424776	Non-hallucinogenic analog reproduces effect without opioid receptors
Mu-opioid antagonism	Complementary (secondary within seed model)	P 26302653 P 24204784	Real but weak (Ke 20 μ M); argues against substitution, not a primary driver
KOR-signalling normalization / antagonism	Competing reinterpretation of same node — opposite polarity	P 26302653 (β -arrestin inhibition), 38649428, 39019223, 32894343	Reframes benefit as functional KOR-signalling dampening; better explains antidepressant effect
NMDA antagonism	Parallel legacy hypothesis	P 26807959	μ M NMDA affinity historically invoked; not independently favored
Non-specific polypharmacology	Umbrella alternative — no single edge necessary	P 26807959 P 9685673	Parsimonious given convergent phenotype

These are not mutually exclusive: ibogaine is polypharmacological, and the effect is plausibly the sum of several μM -affinity actions, with noribogaine shifting the balance toward the opioid/kappa arm over time.

Proposed Follow-up Experiments / Discriminating Tests (ranked by decisiveness)

1. **Kappa necessity test (most decisive).** *Oprk1*-KO/conditional-KO and norBNI pretreatment before ibogaine/noribogaine in rat opioid (heroin/fentanyl) self-administration + opioid-induced hyperalgesia; include both sexes. *If model holds:* anti-opioid effect abolished. *If preserved:* kappa dispensable for ibogaine — refutes necessity while leaving oxa-iboga sufficiency intact.
 2. **Target-engagement / occupancy study.** KOR PET occupancy (e.g., [^{11}C]LY2795050) vs plasma/brain noribogaine across CYP2D6 phenotypes; correlate occupancy with behavioral suppression. Resolves the exposure-gating gap.
 3. **Polarity test.** Compare oxa-noribogaine (biased KOR agonist) vs aticaprant (KOR antagonist) in ICSS and anhedonia assays after chronic opioid/withdrawal. Determines whether anti-reward relief is agonist- or antagonist-like.
 4. **Mechanism-dissection panel.** In matched OUD models, compare oxa-noribogaine (kappa), 18-MC ($\alpha 3\beta 4$), tabernanthalog (5-HT_{2A}) $\pm$ their respective antagonists $\pm$ anti-GDNF; measure VTA GDNF. Partitions the convergent phenotype among routes.
 5. **CYP2D6-stratified human PK/PD RCT.** Genotype-stratified dosing with noribogaine exposure, KOR pharmacodynamic biomarkers (e.g., prolactin), and withdrawal/anhedonia scales.
 6. **Pharmacogenomic association.** Test OPRK1/PDYN and CYP2D6 variants against ibogaine/oxa-iboga response in any available cohort/registry (closes G4).
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Curation Leads (require curator verification)

Candidate status: Retain **EMERGING**; annotate that the model's **sufficiency** claim is supported while its **necessity** claim is untested; flag the direction-of-effect paradox prominently; add restriction that behavioral kappa evidence is male-rat and analog-based.

Candidate evidence references + snippets to verify: -

P 26302653

— "noribogaine was a G-protein biased kappa agonist 75% as efficacious as dynorphin A at stimulating GDP-GTP exchange" (SUPPORT, IN_VITRO; already in KB). -

P 24204784

— "ibogaine, noribogaine and 18-MC were MOR antagonists with functional Ke values ranging from 3 uM (ibogaine) to 13 uM (noribogaine and 18MC)" (QUALIFIES mu claim, IN_VITRO; new). -

P 15659598

— "the ibogaine-mediated decrease in ethanol self-administration was mimicked by intra-VTA microinjection of GDNF and was reduced by intra-VTA delivery of anti-GDNF neutralizing antibodies" (COMPETING, MODEL_ORGANISM, loss-of-function; new). -

P 15178360

— "These data are consistent with the importance of nicotinic alpha3beta4 receptors as a therapeutic target to modulate drug seeking" (COMPETING, MODEL_ORGANISM; new). -

P 33299186

— "The psychedelic alkaloid ibogaine has anti-addictive properties in both humans and animals" (COMPETING context, 5-HT2A/plasticity; new). -

P 28402682

— "Reductions in BDI-II scores from baseline to 12-month follow-up were also significant ($p < 0.001$)" (SUPPORT + QUALIFIES; antidepressant paradox; new). -

P 38649428

— "Improvement ... in MADRS total score at week 6 for aticaprant was significant versus placebo" (COMPETING/direction; new). -

P 23101464

— "Synthetic κ -opioid receptor (KOR) agonists induce dysphoric and pro-depressive effects" (tension against KOR-agonist therapy; new). -

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— "Ibogaine cardiac side effects (QTc time) and cerebellar effects are most likely more driven by ibogaine rather than noribogaine" (QUALIFIES; exposure/CYP2D6; new). -

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— "seven studies ... showed no improvement over controls" (QUALIFIES/negative replications, REVIEW; new).

Candidate pathophysiology nodes/edges: - Node: **noribogaine** —biased_agonist→ **KOR (OPRK1)** [G-protein pathway] (established, in vitro). - Edge: **noribogaine** —functional_antagonist→ **dynorphin/KOR β -arrestin recruitment** (in vitro; label UNCONFIRMED in vivo). - Edge: **KOR engagement** → **↓ opioid self-administration** — mark **UNCONFIRMED (missing perturbation evidence)**. - Competing edges: **ibogaine**→**VTA GDNF**→**↓ drug SA** (LOF-validated, alcohol); **18-MC**→ **$\alpha 3\beta 4$ nAChR**→**↓ drug SA**; **tabernanthalog**→**5-HT2A**→**neuroplasticity**→**↓ seeking**. - Gating node: **CYP2D6 genotype**→**noribogaine exposure**→(gates) target engagement.

Candidate ontology terms: kappa opioid receptor (OPRK1); prodynorphin (PDYN); β -arrestin-2 (ARRB2); GPCR signaling pathway (GO:0007186); $\alpha 3\beta 4$ nicotinic receptor (CHRNA3/CHRNA4); GDNF/RET signaling (GO:0035860); serotonin 5-HT2A receptor (HTR2A); CYP2D6; ventral tegmental area; nucleus accumbens; medial habenula; "drug self-administration"; "hyperalgesia"; "hyperkatifeia/negative affect".

Candidate knowledge_gaps / discussion prompts: (1) "No norBNI/*Oprk1*-knockout test of ibogaine's anti-opioid effect exists — kappa necessity unproven." (2) "Direction-of-effect conflict with KOR-antagonist antidepressant class (aticaprant/navacaprant)." (3) "In vivo KOR occupancy at therapeutic human noribogaine exposure unmeasured." (4) "Negative/failed preclinical replications exist (

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(5) "No OPRK1/PDYN or CYP2D6 pharmacogenomic evidence for ibogaine response retrieved as of 2026-08-28."

Limitations of this Search

Literature-only (no primary datasets analyzed); targeted pharmacogenomic/registry sources (PharmGKB, GWAS Catalog, ClinicalTrials.gov) were not directly queried and are listed as curation to-dos rather than confirmed absences; several key papers were available only as abstracts. The judgment reflects primary in vitro, model-organism, human clinical, and human genetic studies; reviews are labeled as review-level orientation.

Report generated 2026-08-28 from a 5-iteration hypothesis search (35 papers reviewed, 6 findings recorded).

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